Ammal Abbasi

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Education

University of California San Diego **Sep 2019 – June 2025**
Ph.D., Bioinformatics and Systems Biology *La Jolla, CA*

Mount Holyoke College **Sep 2015 – May 2019**
B.A.,magna cum laude, Computer Science and Biology *South Hadley, MA*

Research Appointments and Experience

Postdoctoral Scholar | Alexandrov Lab, UC San Diego **June 2025 – Present**
Principal Investigator: Ludmil B. Alexandrov, Ph.D.

- Developing pksProfiler, a Nextflow pipeline for robust detection and characterization of colibactin-encoding *pks* islands in tumor sequencing data and metagenomes. The pipeline uses sequence alignment and hidden Markov models to assess island completeness, visualize coverage, and identify the carrying bacterial species or strain where the data permit.
- Developed and maintain CMPipeline, a cancer microbiome pipeline for robust microbial detection from sequencing data originally collected to study human tumors. Used the pipeline to lead microbiome analyses of 981 colorectal cancer genomes from 11 countries in the Mutographs study, published in *Nature* (2025), investigating microbial contributions to early-onset colorectal cancer.
- Building a life-course atlas to establish the prevalence and persistence of *pks*-positive bacteria and associated microbial community dynamics across the lifespan in health, precancer, and cancer. Integrating 30,000 cancer genomes from TCGA, Hartwig, Genomics England, and Mutographs with 40,000 infant and adult metagenomes to assess associations with clinical features, age, and environmental exposures.
- Developing an AI-assisted knowledge graph linking microbial exposures, host biology, and clinical and lifestyle data to investigate how their interactions influence colorectal cancer risk and identify priorities for prevention research.
- Developing and benchmarking non-negative matrix factorization (NMF) methods to identify recurring microbial community patterns in cancer sequencing data and assess their reproducibility across TCGA and PCAWG cohorts.
- Developing a machine learning tool to predict platinum and PARP inhibitor response in prostate and pancreatic cancers using whole-genome and whole-exome sequencing data.

Graduate Student Researcher | Alexandrov Lab, UC San Diego **Sep 2019 – June 2025**
Principal Investigator and Thesis Chair: Ludmil B. Alexandrov, Ph.D.
Thesis Co-Chair: Rob Knight, Ph.D.

- Built and patented HRProfiler, a machine learning tool that predicts platinum and PARP inhibitor response in breast and ovarian cancer using WGS/WES data; This work is published in *Cancer Research*, and is available as a Python package.
- Co-authored High-Impact Computational Tools:
  - SigProfilerExtractor: Machine-learning approach for mutational signature extraction (*Cell Genomics*).
  - Copy Number Signatures: Novel input framework for copy number signature extraction (*Nature*).
  - DeepHRD: Deep learning model detecting HRD and predicting platinum response from routine histology slides (*Journal of Clinical Oncology*, 2024).
  - Repeat Sequence Pipeline: Python workflow for palindromic repeat sequence detection in cancers (*Nature*).
- Invited to write review articles on mutational signatures in *Nature* and *DNA Repair*, and selected to present at international conferences including WHO (France), Future Leaders Conference (Spain), and EMBL Conference (Germany).

UCSD Clinical Shadowing | Dr. Rana Mckay's Clinic, UC San Diego Health **Aug 2023 – Dec 2023**

- Shadowed Dr. Rana R. McKay, a leading medical oncologist specializing in prostate and genitourinary cancers, at the Koman Family Outpatient Pavilion. Observed patient consultations, tumor board discussions, and treatment planning for advanced prostate cancer within a multidisciplinary care setting.
- Participated in clinical discussions on integrating genomic testing to guide drug selection and personalize therapeutic strategies.

Computational Biologist | MGH Cancer Center **May 2017 – Sep 2019**

Principal Investigators: Michael S. Lawrence, Ph.D., and Aaron N. Hata, M.D., Ph.D.

- Developed scalable genomic pipelines in MATLAB and Python to analyze patient and patient-derived cell line data, linking mutational signatures, driver genes, and chromosomal instability to therapy-induced APOBEC3A activity in lung cancers. This work is published in *Nature*.

Bioinformatics Scientific Scholar | Novartis Institutes for BioMedical Research **May 2018 – Aug 2018**

- Developed a web-based SQL and RShiny application for querying and visualizing array genotyping data from 17000 patients across 32 clinical trials; This project was awarded best poster for innovation and impact.

Publications

Peer-Reviewed Research Articles

1. Hartwich TMP, Bellone S, De Tommasi O, Ottum S, Ettorre V, Greenman M, Sethi N, Demirkiran C, Yang KY, **Abbasi A**, Alexandrov LB, Santin AD. (2026). Whole-Exome Sequencing Identifies Candidate Genomic Features Associated with Response to Platinum-Based Chemotherapy and Ixabepilone-Based Treatment in Ovarian Cancer. ***International Journal of Molecular Sciences***, 27(17), 7524. doi: 10.3390/ijms27177524.
2. Kazachkova M, Otlu B, Díaz-Gay M, **Abbasi A**, Moody S, Jiang Z, Perdomo S, Wedge DC, Brennan P, Stratton MR, Alexandrov LB. (2026). Identification and validation of a previously missed mutational signature in colorectal cancer. ***Nature Communications***, 17(1), 9671. doi: 10.1038/s41467-026-76472-w.
3. **Abbasi A**, Steele CD, Bergstrom EN, Khandekar A, Farswan A, McKay RR, Pillay N, Alexandrov LB. (2025). HRProfiler Detects Homologous Recombination Deficiency in Breast and Ovarian Cancers Using Whole-Genome and Whole-Exome Sequencing Data. ***Cancer Research***, 85(13), 2504–2513. doi: 10.1158/0008-5472.CAN-24-2639.
4. Díaz-Gay M, dos Santos W, Moody S, Kazachkova M, **Abbasi A**, Steele CD, Vangara R, et al. (2025). Geographic and age variations in mutational processes in colorectal cancer. ***Nature***, 643(8070), 230–240. doi: 10.1038/s41586-025-09025-8.
5. Torrens L, Moody S, de Carvalho AC, Kazachkova M, Abedi-Ardekani B, Cheema S, Senkin S, Cattiaux T, Cortez Cardoso Penha R, Atkins JR, Gaborieau V, Chopard P, Carreira C, **Abbasi A**, et al. (2025). The complexity of tobacco smoke-induced mutagenesis in head and neck cancer. ***Nature Genetics***, 57(4), 884–896. doi: 10.1038/s41588-025-02134-0.
6. Bergstrom EN, **Abbasi A**, Díaz-Gay M, Galland L, Ladoire S, Lippman SM, Alexandrov LB. (2024). Deep Learning Artificial Intelligence Predicts Homologous Recombination Deficiency and Platinum Response From Histologic Slides. ***Journal of Clinical Oncology***, 42(30), 3550–3560. doi: 10.1200/JCO.23.02641.
7. Khandekar A, Vangara R, Barnes M, Díaz-Gay M, **Abbasi A**, Bergstrom EN, Steele CD, Pillay N, Alexandrov LB. (2023). Visualizing and exploring patterns of large mutational events with SigProfilerMatrixGenerator. ***BMC Genomics***, 24, 469. doi: 10.1186/s12864-023-09584-y.
8. Isozaki H, Sakhtemani R, **Abbasi A**, et al. (2023). Therapy-induced APOBEC3A drives evolution of persistent cancer cells. ***Nature***, 620(7973), 393–401. doi: 10.1038/s41586-023-06303-1.
9. Li JSZ, **Abbasi A**, Kim DH, Lippman SM, Alexandrov LB, Cleveland DW. (2023). Chromosomal fragile site breakage by EBV-encoded EBNA1 at clustered repeats. ***Nature***, 616(7957), 504–509. doi: 10.1038/s41586-023-05923-x.

10. Islam SMA, Díaz-Gay M, Wu Y, Barnes M, Vangara R, Bergstrom EN, He Y, Vella M, Wang J, Teague JW, Clapham P, Moody S, Senkin S, Li YR, Riva L, Zhang T, Gruber AJ, Steele CD, Otlu B, Khandekar A, **Abbasi A**, et al. (2022). Uncovering novel mutational signatures by de novo extraction with SigProfilerExtractor. ***Cell Genomics***, 2(11), 100179. doi: 10.1016/j.xgen.2022.100179.
11. Steele CD, **Abbasi A**, Islam SMA, et al. (2022). Signatures of copy number alterations in human cancer. ***Nature***, 606(7916), 984–991. doi: 10.1038/s41586-022-04738-6.

Reviews and Commentaries

1. **Abbasi A**, Alexandrov LB. (2026). Biological aging and the hidden architecture of early-onset cancer risk. ***Cancer Cell***, 44(9), 1743–1745. [Spotlight] doi: 10.1016/j.ccell.2026.07.018.
2. **Abbasi A**, Alexandrov LB. (2022). Unwinding the mutational signatures of a DNA topoisomerase enzyme. ***Nature***, 602, 580–581. [News and Views] doi: 10.1038/d41586-022-00301-5.
3. **Abbasi A**, Alexandrov LB. (2021). Significance and limitations of the use of next-generation sequencing technologies for detecting mutational signatures. ***DNA Repair***, 107, 103200. [Review] doi: 10.1016/j.dnarep.2021.103200.

Preprints

1. Di Ciccio S, van Strien J, Jiang Y, **Abbasi A**, Bakker L, Weertman N, Leiteris L, Willems J, Alexandrov LB, Garaycoechea JI. (2026). Mutagenic bypass of uracil-derived abasic sites underlies the SBS17 mutational signature. ***bioRxiv***. [Preprint] doi: 10.64898/2026.08.21.746158.
2. Abbasi E, **Abbasi A**, Akçay E. (2025). Competition, Mutualism, and Host Immune Control in a Cancer Microbiome. ***bioRxiv***. [Preprint] doi: 10.1101/2025.05.06.652496.

Manuscripts in Preparation

1. **Abbasi A**, et al. The colorectal cancer microbiome across 11 countries. Manuscript in preparation.
2. **Abbasi A**, et al. pksProfiler detects colibactin biosynthetic islands in cancer genomes and metagenomes. Manuscript in preparation.
3. **Abbasi A**, et al. A life-course atlas of colibactin-encoding bacteria in health and disease. Manuscript in preparation.

Selected Conference Abstracts

1. **Abbasi A**. (2026). The role of pks-positive bacteria in colorectal cancer. ***Cancer Research***, 86(8 Suppl), LB208. [Conference abstract] doi: 10.1158/1538-7445.AM2026-LB208.
2. Kazachkova M, Cattiaux T, **Abbasi A**, Meyenberg M, Loizou J, Stratton MR, Brennan P, Alexandrov LB. (2026). Mutational signatures associated with Western dietary patterns in colorectal cancer. ***Cancer Research***, 86(8 Suppl), LB164. [Conference abstract] doi: 10.1158/1538-7445.AM2026-LB164.
3. Madakamutil Y, Joseph L, Rahman D, **Abbasi A**, Alexandrov L. (2026). Leveraging transcriptomic profiles and deep learning to detect homologous recombination deficiency in breast cancer with softHRD. ***Cancer Research***, 86(7 Suppl), 1046. [Conference abstract] doi: 10.1158/1538-7445.AM2026-1046.
4. **Abbasi A**, Alexandrov L. (2024). The microbial landscape of colorectal cancer across 11 countries. ***Cancer Research***, 84(7 Suppl), LB374. [Conference abstract] doi: 10.1158/1538-7445.AM2024-LB374.
5. Bergstrom EN, **Abbasi A**, Díaz-Gay M, Galland L, Ladoire S, Lippman SM, Alexandrov LB. (2024). Deep learning AI predicts HRD and platinum response from histologic slides. ***Cancer Research***, 84(7 Suppl), LB391. [Conference abstract] doi: 10.1158/1538-7445.AM2024-LB391.

Selected Presentations

2025 SoCal Genome Stability Symposium | City of Hope, USA **Dec 2025**
Invited Speaker; HRProfiler Detects HRD in Breast and Ovarian Cancers Using WGS and WES Data.

AACR Annual Meeting | San Diego, USA **April 2024**
Poster Presenter; Microbial Landscape of Colorectal Cancers Across 11 Countries.

EMBL Cancer Genomics Conference | Heidelberg, Germany **November 2023**
Poster Presenter; HRProfiler Detects HRD in Breast and Ovarian Cancers Using WGS and WES Data.

SoCal Genome Stability Symposium | Irvine, USA **May 2023**
Poster Presenter; HRProfiler Detects HRD in Breast and Ovarian Cancers Using WGS and WES Data.

Skype a Scientist | Virtual **March 2023**
Invited Speaker; "Roadmap of My Scientific Career" talk for Fremont Union High School AP students

Cancer Grand Challenges Future Leaders Conference | Barcelona, Spain **November 2022**
Poster Presenter; HRProfiler Detects HRD in Breast and Ovarian Cancers Using WGS and WES Data.

Teaching Experience

University of California San Diego **Spring 2023**
Teaching Assistant, Statistical Learning in Bioinformatics *La Jolla, CA*

- Developed course materials, led active-learning sessions, and provided feedback on assignments and project presentations involving statistical learning and analysis of biological datasets.

University of California San Diego **Spring 2021**
Teaching Assistant, BENG 100 *La Jolla, CA*
Introduction to Probability and Statistics for Bioengineers.

- Led a weekly discussion section explaining worked problems in probability and statistics, including probability distributions, statistical estimation, and hypothesis testing for bioengineering applications.

Research Mentoring

University of California San Diego **Jan 2021 – Present**

Master's and undergraduate research

Mentor students developing transcriptome-based methods to detect homologous recombination deficiency (HRD), including softHRD.

- Four master's and undergraduate mentees; one is now a Ph.D. student at MD Anderson Cancer Center and one works in industry.

Research output: softHRD poster, AACR Annual Meeting 2026 (Abstract 1046), presented by my mentees.

High school research

Guide analyses of colorectal cancer genomes to identify and interpret mutational patterns.

- Three high school students.

Leadership and Community Engagement

Co-Vice Chair of Inclusion and Belonging **2025 – 2026**

UC San Diego Postdoctoral Association

- Co-led initiatives to strengthen inclusion, community, and peer support for postdoctoral scholars through events and opportunities for connection and collaboration.

MicroMBA | UCSD **July 2023 - August 2023**

- Successfully completed coursework covering key business concepts such as operational strategy, finance, and entrepreneurship while engaging in networking sessions with industry professionals and STEM graduate students.

Graduate Community Assistant | UCSD **March 2023 – May 2025**

- Planned, organized, and hosted multiple community-building events tailored to the diverse needs of over 500 graduate student residents, fostering an inclusive and supportive living environment.
- Collaborated with Housing and Residential Life to identify community interests and design programming that promoted well-being, engagement, and academic-life balance.

Invited Scientific Speaker | Skype a Scientist Org. **March 2023**

- Delivered a presentation to AP students at Fremont Union High School, detailing my scientific trajectory, career choices, and offering valuable advice for those aspiring to pursue graduate studies.

Patent Applications

Alexandrov LB, **Abbasi A**. Methods and Systems for Detecting Homologous Recombination Deficiency in Cancer Therapies. International patent application PCT/US2023/068465; publication WO2023245082A2. Filed June 14, 2023; published December 21, 2023.

Technical Skills

- **Programming & Data Analysis:** Python (Pandas, NumPy, Matplotlib), R (ggplot2, Shiny), Bash, Linux; Java, C++; Machine Learning (scikit-learn); Deep Learning (TensorFlow, PyTorch); NLP & LLMs (LangChain); Knowledge Graphs (Neo4j).
- **Bioinformatics & Cancer Genomics:** NGS analysis (WGS, WES, RNA-seq, long-read, and metagenomics); variant calling and annotation (SNVs, CNAs, Structural Variants); mutational signature analysis; pipelines/workflows (GATK, QIIME2, Nextflow, Snakemake).
- **Infrastructure & Tools:** HPC and Slurm; containerization (Docker, Singularity); cloud computing (AWS S3, EC2); version control (Git); interactive computing (Jupyter Notebooks).